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Power system frequency profile improvement using satin bowerbird optimized tilt fractional cascade controller with reduced data based cost function

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ABSTRACT

The objectives of the automatic generation control (AGC) are attained via secondary control mechanism with optimal controller gains. The best solutions of these gains are identified by population search-based algorithms with the help of performance metrices. The controller design issues of AGC treat these performance metrics as a cost function. This paper introduces a new performance measure implemented with dominated samples information extracted from the frequency and tie line power changes. This new integral square error (NISE) is incorporated in fitness function and tested on isolated and multi area interconnected power system scenarios with satin bowerbird optimizer to tune the controller gains. Additionally, integral-tilt proportional-fractional order derivative controller is proposed in this paper and comparisons are made with popular controllers to show the merits in terms of AGC specifications. Optimal gain parameters of all controllers are achieved with existing integral square error and the proposed NISE cost functions and extensive simulations are carried out to illustrate the merits of the work. Studies are also expanded to nonlinear systems in order to evaluate the adaptability, performance improvement, and effectiveness of the suggested cost function, controller, and tuning method.

Introduction

Motivation and incitement

In a power system, the frequency is monitored and continuously regulated by the primary and secondary control loops during power mismatches caused from load variations. The AGC mechanism incorporated in the system, controls the plants inputs to meet the demand and to reduce the frequency and inter area tie-line power disturbances. Power generation is maintained by primary control action via regulation constant to catch the demand. The frequency profile of the system is further enhanced using the secondary control action. Among these control schemes, extensive research is carried out on the secondary controller selection, design, and optimization [1–4]. To minimize the frequency deviations of the isolated and interconnected power system using these secondary controllers, optimal gain values are needed. In this context, population based heuristic and intelligent algorithms are helpful with suitable fitness and/or cost functions. The cost functions are designed with frequency information, and tie line power deviations since the objective of the AGC is to minimize these deviations. IAE, ITAE, ISE and ITSE are the performance indices used by earlier researchers to find optimal parameter gains of the supplementary controllers.

Literature review

Various models of the power system were investigated in past to incorporate AGC mechanism. Majority works used linearized models of isolated/two/three area interconnected systems [1]. Later, conventional systems with boiler dynamics, governor dead band, generation rate constraint and parameter uncertainties were considered to study their impact on AGC. The main objective of the AGC is to minimize the frequency and tie line power deviations irrespective of system model. The LFC strategies are another pioneer work in this area. A detailed review of all classical and advanced control techniques available for AGC are provided in [2]. In smart grid environment, the impact of

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Nomenclature

Δf_i	Change in frequency of i^{th}area.
ACE_i	Area control error of i^{th}area.
2-DOF	2-degree of freedom.
AGC	Automatic generation control.
AVR	Automatic voltage regulator.
FACTS	Flexible AC Transmission Systems.
FO	Fractional order.
IAE	Integral absolute error.
ISE	Integral square error.
ITAE	Integral time absolute error.
ITSE	Integral time square error.
I-TPFOD	Integral-tilt proportional-fractional order derivative.
LFC	Load frequency control.
NISE	New integral square error.
PID	Proportional integral derivative.
SBO	Satin bowerbird optimizer.

renewables and energy storage elements on AGC is challenging and significant works are reported in this aspect to meet the objectives [3]. In conventional system, PID controller is popular as supplementary controller and challenges of design of PID controller in AGC are reported in [4]. Further, FOPID controllers are proposed to enhance the control action of the secondary control loop [5]. In AGC perspective, detailed reviews are presented from conventional to future smart grid in [1–6]. Precisely, isolated power system frequency control is presented in [7]. In [7], the mutual effects of LFC and AVR in presence of storage units with traditional PI and PID as supplementary controllers are considered. AGC controllers applied for multi-source and multi area power systems [8,9] and extended to modern power systems scenarios include renewable energy sources and energy storage units [10]. Majority of these papers are based on single objective frameworks [11] to identify the controller parameters. Multi-objective optimization approaches are also tested in AGC studies to enhance the stability of various models in diverse conditions [12–14]. The works reported in [7–14] used PI and PID as secondary controllers. To improve the frequency and tie-line power profiles, cascade control schemes are also applied in recent works to get superior results than PID controller. In [15], multistage controller is suggested for interconnected power system and PI-PD controller is used in [16] for similar system in deregulated environment. Second order derivative component is added with PID to boost the performance of the secondary controller of AGC in multi area power system is reported in [17]. Weighted PID controllers are also available in the literature in frequency control studies. 2-DOF-PID controller and 2-DOF-FOPID controllers were investigated in [18], [19] for AGC applications. The optimal values of the suggested controllers are searched by heuristic algorithms such as particle swarm optimizer, differential evolution, BAT algorithm, invasive weed optimizer etc. On the second phase, intelligent controllers like fuzzy [20,21], sliding mode control [22], model predictive control [23] are extended to suit AGC.

Recently variants of fractional and tilt controllers are applied in AGC under different scenarios with more realistic models [24,25]. These controllers produce enhanced AGC responses in terms of peak overshoot, settling time and steady state error over traditional PID controllers. Therefore, combination of fractional and tilt controller may provide such results. There are a few of these common AGC studies using fractional order controllers in the literature [26–29]. The FOPID variations outperformed traditional controllers in the presence of FACTS [26]. Additionally, in AGC research [27,28], fuzzy and neuro fuzzy assisted fractional order controllers improve the secondary controller's decision-making process. Design of the secondary controller in presence of high renewable energy penetration [30], optimal-dispatch based AGC [31,32] and optimal mileage-based AGC in presence of renewables [33] are some new emerging ideas to meet the objectives of the AGC without sacrificing the fuel cost benefits. Apart from controller's design, communication failures, cyber-attacks, time delays and intentional false data injections are severe problems in smart modern power system [34,35]. These challenges are addressed in limited works. All these sub domains of AGC required cost or fitness functions since controller tuning is necessary task in each problem [36] and optimal values of controllers are achieved with the help of cost functions. Few new performance measures are suggested in literature [37,38] along with existing measures. In this paper, this area is focused along with other challenges.

In addition to the comprehensive studies on the design, tuning, and application of supplemental controllers, there are few works that have been done to pinpoint the new performance indices for AGC. This is required to enhance the frequency and tie line power profiles of the isolated and interconnected power systems at different operating conditions and configurations. Recent research [37,38] presented the peak values of the frequency and tie line disturbance information considered in the performance indices and new cost functions. These indices utilized peak values of the disturbed signals along with existing indices. However, these new indices require more sample than existing indices which can further increase computational burden and memory requirement. Although improvements were observed compared to ISE, ITSE, IAE and ITAE assisted schemes, but still requires additional algorithm known as 'peak detection'. The usage of these peak values enhances the cost function mechanism. This idea of peak samples usage in objective function calculation motivated authors to design new performance measure. The main objectives of the proposed implementation algorithm are.

- To improve the frequency and tie line profiles irrespective of the controller, tuning algorithm, and system category.
- To reduce the number of samples used in cost function without sacrificing benefits achieved from traditional cost functions.
- To enhance the adaptivity of the mechanism in different scenarios to meet the goals of the AGC.

Contribution

In this work,

- A new performance measure is implemented as cost/fitness function for the tuning of controller which provides improved system outputs during load perturbations. The simulation results are carried out for both isolated and interconnected power system to draw the merits of the method. Comparisons are provided with most acceptable ISE performance index at different loadings, sampling rates and other conditions.
- I-TPFOD controller is proposed as supplementary controller in AGC to enhance the frequency profile of the isolated power system and power flows profile of the tie line in the interconnected power system.
- The gains of controllers are achieved with SBO using ISE and proposed performance index as cost functions. Comparisons with existing controllers provide benefits of the study in terms of peak overshoot, setting time and fitness values.
- In addition, to assess the benefits of the suggested research effort, the studies are extended to an interconnected power system that integrates a significant proportion of renewable energy.

The rest of the article organises as follows: Section 2 provides new cost function information along with existing metrices. Section 3

provides details on the planned I-TPFOD controller. The SBO algorithm is introduced in Section 4 before test systems, simulation results, comparisons, and extensions are presented in Section 5. Section 6 contains the conclusions.

Proposed cost function of AGC

The cost functions used in AGC to find the optimal parameter gains of the secondary/supplementary controllers are framed with the information of individual area frequencies and the tie line powers. These cost functions are affected by changes in frequency profile data in isolated power system models. Area frequencies and tie line power flows of connected regions are used to define the cost functions in interconnected power systems models. The four existing cost functions used by earlier research studies are given by

$$J_{\mathrm{ISE}} = \int_0^t \sum \{(\Delta f_i)^2 + (\Delta P_{ij})^2\} dt \tag{1}$$

$$J_{\mathrm{ITSE}} = \int_0^t t[\sum \{(\Delta f_i)^2 + (\Delta P_{ij})^2\}] dt \tag{2}$$

$$J_{\mathrm{IAE}} = \int_0^t \sum (|\Delta f_i| + |\Delta P_{ij}|) dt \tag{3}$$

$$J_{\mathrm{ITAE}} = \int_0^t t\{\sum (|\Delta f_i| + |\Delta P_{ij}|)\} dt \tag{4}$$

From Eqs. (1)–(4), *i*and*j* depends on the total number of areas. The cost functions of the metrices are represented by J_{ISE}, J_{ITSE}, J_{IAE} and J_{ITAE}. Among these cost functions, preference is given to J_{ISE} in literature [11–22] since it produces relatively best solutions to AGC problems. In discrete data, the cost function is exactly calculated and therefore controller design problem practically valid according to the Eq. (5).

$$J_{\mathrm{ISE}} = \sum_{k=1}^{N} \sum \{(\Delta f_i)^2 + (\Delta P_{ij})^2\} \tag{5}$$

N is the total number of samples in change in frequency and tie line power signals based on the simulation time(t). Few other cost functions are also tried in later stage along with the existing performance indices and the details are available in [37,38]. Especially the peak detection assisted indices helpful to enhance the profiles of frequency and tie line powers of the system when they added with existing indices. This task increases usage of number of samples for calculation of cost function and computational burden. In addition, it require peak detection algorithm to find the peak values of the disturbed signals. To eliminate these drawbacks, a new performance measure is designed in this paper. The proposed new cost function utilize only the dominant samples of the output signal. These dominant samples includes peak samples and adjacent samples of peak values as shown in Fig. 1a. along with state information in Fig. 1b. For the signal δX (change in signal X), the existing ISE is defined as

$$ISE = \sum_{k=1}^{N} \delta X(k)^2 \tag{6}$$

To extract dominant samples from the δX, the $\delta \dot{X}$ is calculated. When this $\delta \dot{X}$ is either zero or close to zero, then corresponding values of δX are considered in the cost function evaluation. Mathematically, $\delta \dot{X}$ represents derivative of δX termed as supported signal and whose values are approaching zero when signal attain either positive peak or negative peak. This concept not only includes peak samples of the signal but also values nearer to peaks. Therefore, the new δX is mathematically represented using Eq. (7).

$$\delta X_{\mathrm{new}}(k) = \begin{Bmatrix} \delta X(k), -\varepsilon \le \delta \dot{X} \le \varepsilon \\ 0, \text{ otherwise} \end{Bmatrix} \tag{7}$$

The new signal δX_{new} consist of a smaller number of non-zero elements which influence cost function value in greater extent. These non-zero elements and fitness function values are depended by ε (threshold parameter). Incorporating this concept in the AGC, the final ISE cost function will be

$$J_{\mathrm{NISE}} = \sum_{k=1}^{M} \sum \{(\Delta f_i)^2 + (\Delta P_{ij})^2\} \tag{8}$$

In Eq. (8), M is the total number of data points of new information whose value is less than *N*. This modified fitness function yields similar and/or superior results than existing performance indices without increasing memory, time, and computational burden.

Integral-tilt-proportional-fractional order derivative (I-TPFOD) controller

The cost or fitness functions are the measures to find the optimal parameter gains of the controllers used in AGC loops. Integral, PI and PID controllers are popular in LFC and AVR studies. Application of other controllers like intelligent, cascade and fractional types are also brought attention to the researchers in AGC related studies. This paper introduced a new controller consist basic components of the PID with additional cascade and fractional mechanisms. The controller consists of two blocks known as integral and tilt-proportional-fractional order derivative component connected in cascade with additional signal support. Fig. 2 shows schematic diagram of the proposed controller.

This structure is applicable for all multi-input and multi-output systems. In AGC application point of view, this structure directly suitable for interconnected power system models since these models have two control input signals *ACE* and Δf. The output of the proposed control is given by

$$c_i(s) = \left(\frac{k_{pi}}{s^{1/n}} + s^{\beta} k_{di}\right)\left(ACE_i(s)\frac{k_{ii}}{s} - \Delta f_i(s)\right) \tag{9}$$

The merits of the fractional order and tilt controls are included using TPFOD block to enhance the stability of the plant in presence of this proposed controller. This structure is modified with additional gain block connected in cascade with integral component to suit for single-input-single-output systems and isolated power system models.

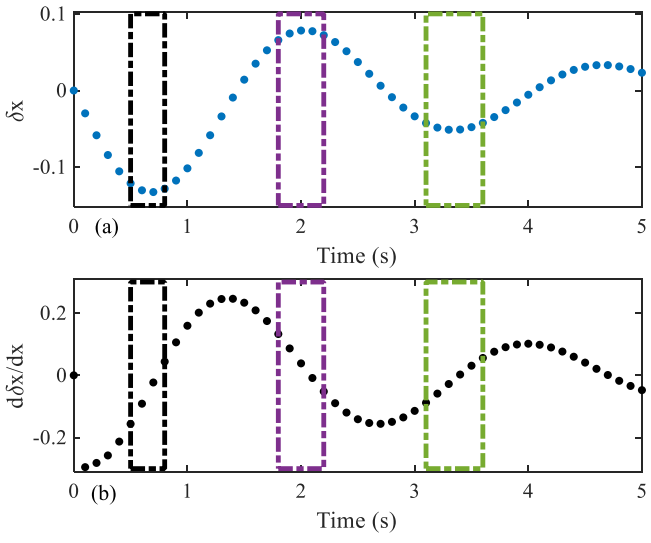


Fig. 1. (a). Signal with dominant samples segregation, (b) state information.

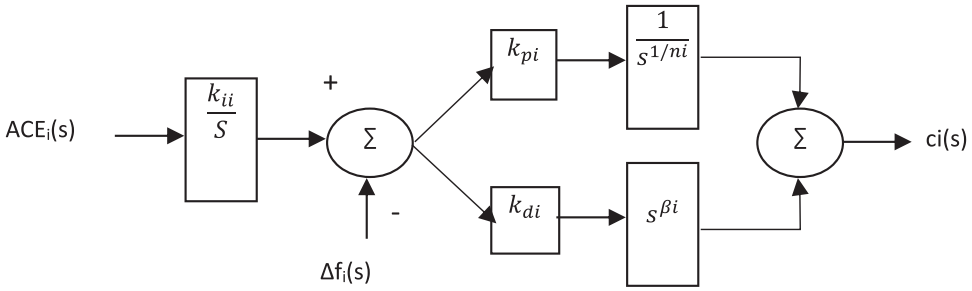


Fig. 2. Proposed I-TPFOD controller applicable for AGC studies.

Overview of satin bowerbird optimizer

Finding the best solution of the controller gains to minimize the frequency and tie-line power deviations of an interconnected power system is an optimization problem. These controller gains are the decision variables evaluated by any heuristic and population search-based techniques. Earlier, ISE, ITSE, IAE and ITAE performance measures are treated as cost/fitness functions of this AGC problem to find the controllers gains. In this paper, the traditional cost functions are replaced by proposed NISE and comparisons are made with ISE cost function. With ISE and NISE cost functions, optimal parameter gains of the supplementary controllers are identified by using SBO algorithm. SBO is heuristic algorithm proposed recently in 2017 by S.H. Moosavi and V.Bardsiri to solve software development effort estimation problem. In this paper, it is extended to solve AGC problem. This algorithm is implemented based on searching, courtship and attracting behaviours of satin bowerbirds. There are male and female bowerbirds distinguished by eyes and colour. The information related to these species is available in [39–41]. By mimic the nature and principles of stain bower bird life into a group of steps, SBO algorithm is designed. According SBO, the stages to solve AGC problem are as follows [40]. The algorithm need the basic information related to common and control parameters to start the process of generating initial solutions. These parameters includes number of bowerbirds (*NB*), number of maximum iterations, mutation probability, spread factor and greatest step size (α). In addition, the number of decision variable along with their upper and lower limits are also predefined before generating initial a set of random bower. Initial solution of the problem generated randomly within the boundaries of the solution space of n-dimensional (depends on the number of decision variables *'n'*). Each solution represents position of bowerbird and corresponding fitness is calculated with the help of cost function. The objective is to minimize the cost function and therefore a best solution is identified from all the solution for next stages. Female bowerbird attracted by bower based on the probability since it is the factor represents the attractiveness of a bower. The probability of individuals is calculating by using Eq. (10).

$$Prob_i = \frac{\text{fit}_i}{\sum_{p=1}^{NB} \text{fit}_p} \tag{10}$$

In Eq. (10), fit_i is the cost function value evaluated for each individual particle. Based on objective function, fit_i will be defined. The best solution in each iteration is the solution corresponding to minimum cost in that iteration and known as elite of the iteration. Elitism allows this mechanism in optimization process. Once best solution is available in a group, it is preserved and used for updating other positions of bowerbirds in later stages. Using Eq. (11) expressed with best and current positions of bowers, new changes are calculated. The Eq. (11) is given by

$$x_{ik}^{\text{new}} = x_{ik}^{\text{old}} + \lambda_k \left(\left(\frac{x_{jk} + x_{elite,k}}{2} \right) - x_{ik}^{\text{old}} \right) \tag{11}$$

In equation (11), x_i is i^{th} bower or solution vector and k denotes decision variable routine. Using roulette wheel selection, j value is selected. x_{elite} indicates elite (best) position. Further, $\lambda_k$is the attraction power calculated using Eq. (12) given by

$$\lambda_k = \frac{\alpha}{1 + p_j} \tag{12}$$

Probability (p_j) and greatest step size (α) are already defined in earlier stages. Using normal distribution, the old positions are updated using the Eq. (13).

$$x_{ik}^{\text{new}} \sim N(x_{ik}^{\text{old}}, \sigma^2) = x_{ik}^{\text{old}} + (\sigma^* N(0, 1)) \tag{13}$$

In Eq. (13), σ is the proportion of space width calculated from upper and lower limits of variables. Once the new solutions are available, the process is repeating like other heuristic algorithm until best solution is identified based on termination rule. The related information of SBO is available in [39]- [41] with different applications. In this paper it is used to find the optimal gains of the AGC controllers. The NISE cost function expressed in Eq. (8) of AGC is used as fitness function of the bowerbirds of SBO with I-TPFOD controller gains as decision variables. The objective of the AGC problem is defined as

$$Z = \min\{J_{\text{NISE}}\} = \min\{ \sum_{k=1}^{M} \sum \{(\Delta f_i)^2 + (\Delta P_{ij})^2\}\} \tag{14}$$

The decision variables are generated by SBO with the help of upper and lower limits defined as follows:

$$k_{pi}^{\min} \le k_{pi} \le k_{pi}^{\max} \tag{15}$$

$$k_{ii}^{\min} \le k_{ii} \le k_{ii}^{\max} \tag{16}$$

$$k_{di}^{\min} \le k_{di} \le k_{di}^{\max} \tag{17}$$

$$n_i^{\min} \le n_i \le n_i^{\max} \tag{18}$$

$$\beta_i^{\min} \le \beta_i \le \beta_i^{\max} \tag{19}$$

From Eqs. (15) to (19), k_{pi}, $k_{ii}$and k_{di} is the proportional, integral, and derivative gains of the ith area. Instead of taking tilt component n_i, $-1/n_i$ is taken as decision variable are replaced Eq. (18) extreme limits.

Simulation studies

To test the possibilities, merits, and limitations of the proposed cost function, two standard test systems are considered. First, simulation studies are carried out on a single area power system and later extended to a multi area interconnected power system. The block diagrams of both test systems are shown in Fig. 3.a and 3.b. The systems parameters are available in appendix 1. Initially, the proposed cost function is tested on the isolated power system without supplementary controller, with integral controller and comparisons are made with ISE cost function. After concluding the merits with conventional studies, I-TPFOD controllers optimal gains are identified by using SBO with NISE cost function. For comparisons in controller perspective, integral and PID controllers are opted. Later, the proposed cost function and controller

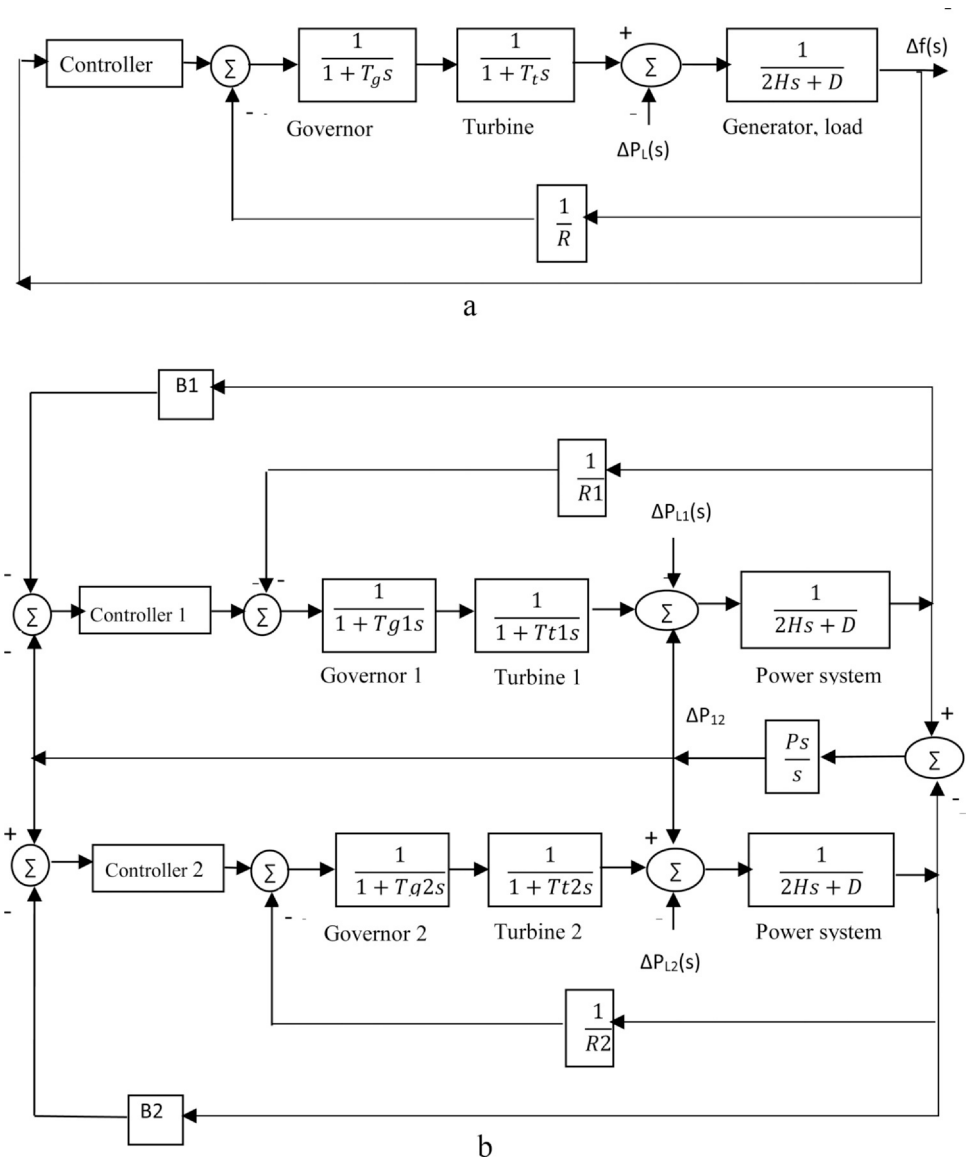


Fig. 3. **a** Isolated single area power system model-test system-1. **b** Interconnected multi-area power system model-test system-2.

are applied for multi area power system. In both cases, SBO algorithm is used to find optimal controllers gains with ISE and NISE fitness function. The parameters of SBO are reported in appendix 2.

Proposed performance measure calculation in case of isolated power system without supplementary controller

The cost functions are necessary to find the optimal values of the controller decision variables. Irrespective of the types of the supplementary controllers, their optimal gains are decided by the search-based algorithms using these cost functions. Before applying the proposed performance measure as cost function of controller problem, it is investigated on isolated system without controller. For this purpose, test system 1 is considered whose Δf and $\Delta \dot{f}$ information is presented in Fig. 4. For this information of frequency changes, existing ISE value is 27.898 utilizing 2000 samples of data with simulation time slot of 20 s. However, samples are reduced to 1847 with new index attain a value of 25.66 for the same frequency profile available in Fig. 5. In this case, the threshold parameter is set as 0.03. Therefore, validated results are guaranteed with proposed cost function with a smaller number of samples once it is incorporated in the controller design.

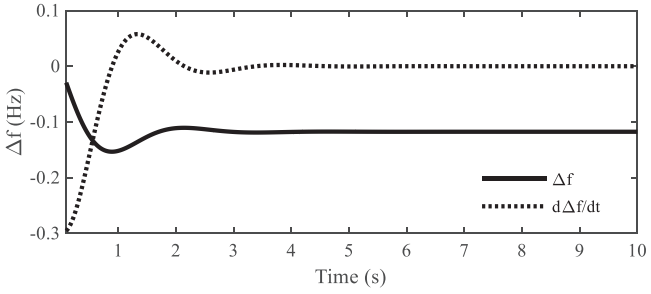


Fig. 4. Change in frequency and supported signals of isolated power system without supplementary signals.

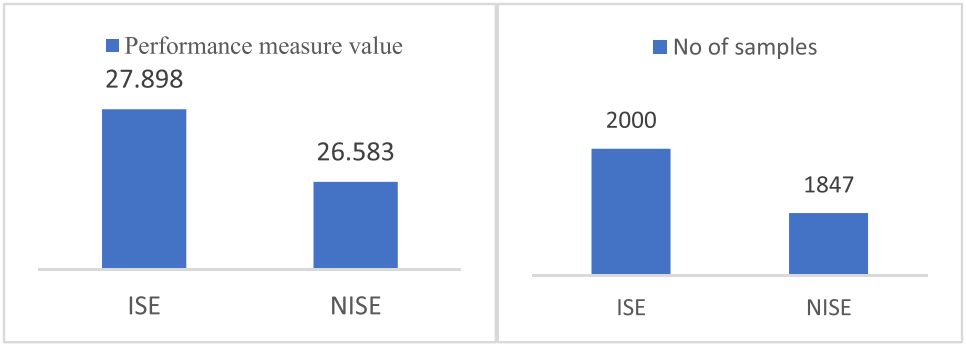


Fig. 5. Performance measures and number of samples information for both ISE and NISE.

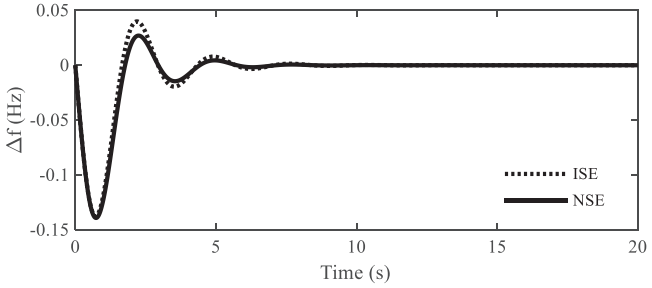


Fig. 6. Δf of isolated power system with integral controller optimized by SBO with ISE and NISE cost functions.

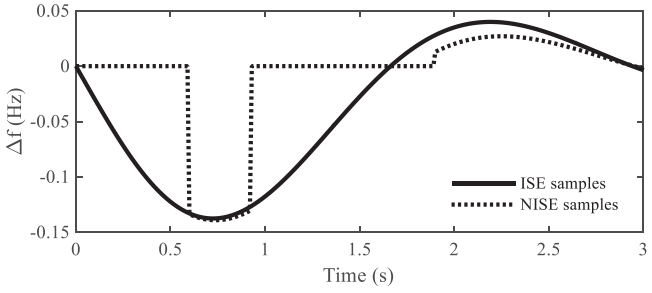


Fig. 7. Samples used by ISE and NISE cost functions.

Proposed performance measure calculation in case of isolated power system with integral controller

The optimal gain value of integral controller of isolated power system can be identified using SBO with the help of both ISE and NISE cost functions. These cost functions values are varying with signal sampling rate. At sampling interval of 0.01, the ISE cost function value is 1.6151 and controller gain is − 0.41748. The cost function and integral controller gain values are 0.6568 and − 0.36171. The change in frequency of the test system at 5% load decreased case is shown in Fig. 6 for both performance measures.

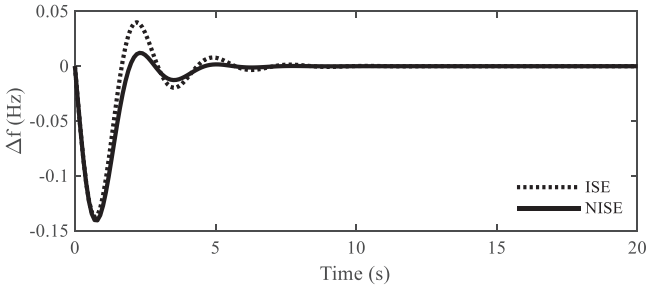


Fig. 8. Isolated system frequency change in case of 0.1 sampling interval.

For the response achieved with ISE, cost function value is 1.6152. In case of NISE, cost function value of the improved response is 0.67393. To validate the approach, NISE and ISE values are calculated for the signals achieved with ISE and NISE cost functions respectively. These values are 0.69138 and 1.6385 relatively larger than optimal cost values. Further, 2000 samples are utilized by ISE cost function and 1845 samples by NISE to produce optimal integral gain value through SBO. For all these results, threshold parameter is 0.08745. Although, a smaller number of samples are used in NISE produced superior results compared to ISE as reported in Fig. 6. These samples information for both ISE and NISE cost functions are plotted in Fig. 7.

Effect of sampling interval of signals on new performance measure

In case of ISE cost function, all the samples of Δf are participated to estimate its value. At different sampling intervals, these measures will change. Similar change is observed in case of proposed performance measure. However, frequency profile enhancement will carry with sampling interval changes. To justify this case study, three sampling intervals related simulations are carried out for both ISE and NISE fitness functions. The sampling intervals are 1 s, 0.1 s and 0.01 s and corresponding integral gain values are reported in Table 1.

Fig. 8 shows the comparison of single area power system frequency changes with integral controller whose optimal parameter gains are identified by SBO with ISE and NISE cost functions. With 186 samples

Table 1
Controller, cost function values of isolated power system at different sampling intervals of the test signal (simulation time of 60 s).

Parameter	ISE			NISE		
	$T_s = 1$	$T_s = 0.1$	$T_s = 0.01$	$T_s = 1$	$T_s = 0.1$	$T_s = 0.01$
Controller gains	-0.40922	-0.41748	-0.41748	-0.29937	-0.30346	-0.36171

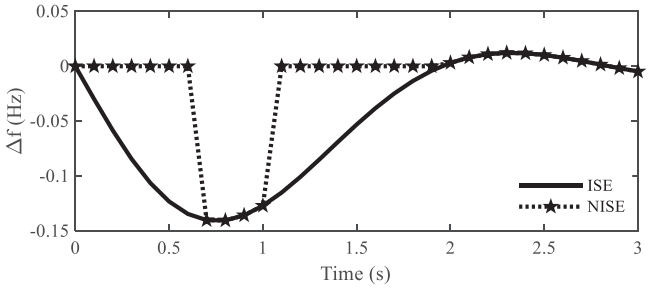


Fig. 9. Samples tracked from signal for ISE and NISE cost functions at Ts = 0.1.

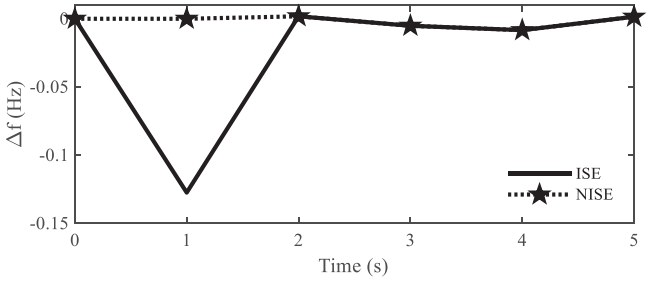


Fig. 10. Samples tracked from signal for ISE and NISE cost functions at Ts = 1.

(at 0.1 sampling interval), NISE cost function yield superior results than ISE evaluated for 200 samples. These samples are plotted in Fig. 9. For Δf of both plots, ISE values are 0.16193, 0.17241 and NISE values are 0.083922, 0.075919. This result shows performance index design significance in AGC along with optimal gains tuning. Similar simulations are carried out at sampling interval of 1 s, and controller gains are provided in Table 1. The samples tracking for calculation of ISE and NISE are presented in Fig. 10.

Observations at different loadings

The proposed cost function is applied at different loadings to find the optimal parameter gains of the controller. These observations are made with integral controller and extended further to other controllers. For different load changes, the optimal parameter gains of integral controller are presented in Table 2. In each case, the responses are improved in terms of peak overshoot and settling time.

Proposed controller optimal gain parameters using new cost function

From the extensive case studies investigated on isolated power system at different sampling intervals, loadings with proposed cost function and comparisons with ISE in terms of fitness values, number of samples and improvements in the AGC mechanism, this new cost function is set to be ready for all other supplementary controllers available in literature. In this paper, authors introduced I-TPFOD whose optimal parameter gains are achieved with SBO algorithm. The architecture of the proposed controller for isolated systems is presented in Section 3. The optimal gain values are

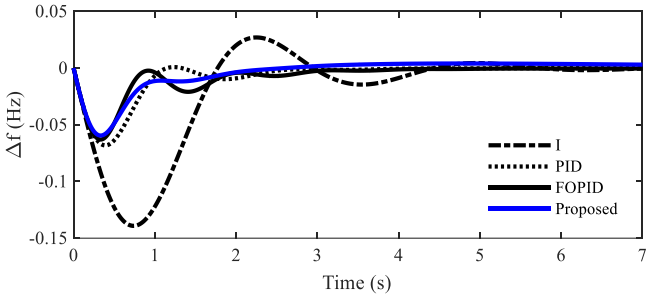


Fig. 11. Comparison of proposed controller with traditional controllers in case of single area power system.

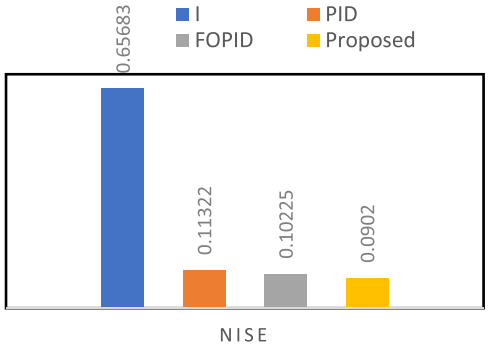


Fig. 12. NISE cost function values for different controllers.

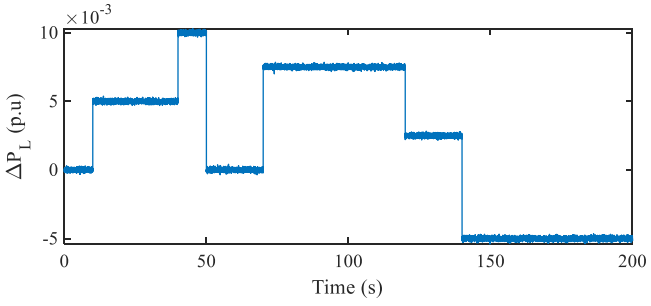


Fig. 13. Random load perturbations simulation result.

$k_p = 0.8784$, $k_i = -0.70016$, $k_d = 0.45144$, $n = 0.45144$, $\beta = 0.6624$ *and* $k = 0.86139$. The frequency changes of the single area power system in presence of proposed controller are presented in Fig. 11.

The results in Fig. 11 shows the improvements when the traditional controllers are replaced by proposed I-TPFOD controller. As proposed controller implemented with fractional calculus, FOPID controller is also used in comparison purpose for this case. It is clearly observable from the Fig. 11 that the peak overshoot, settling time and number of oscillations are reduced with proposed controller. Once the response improved, the number of samples utilized for NISE cost function are reduced and corresponding values are presented in Fig. 12.

Table 2
Controller gains at different loadings with ISE and NISE cost functions.

Cost function	$\Delta P_L = 0.01$	$\Delta P_L = -0.01$	$\Delta P_L = 0.05$	$\Delta P_L = -0.05$
ISE	-0.4203	-0.4203	-0.4175	-0.4175
NISE	-0.4084	-0.4084	-0.3617	-0.3617

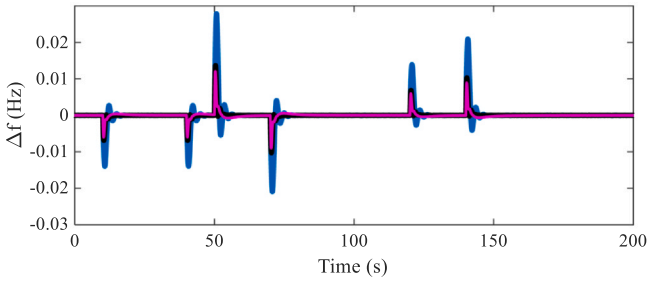


Fig. 14. Change in frequency results of isolated power system at random loadings.

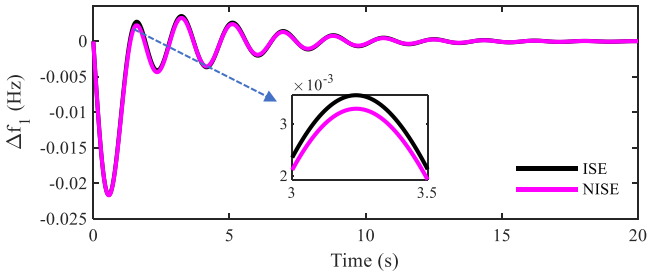


Fig. 15. Interconnected power system area 1 frequency deviations with two cost functions.

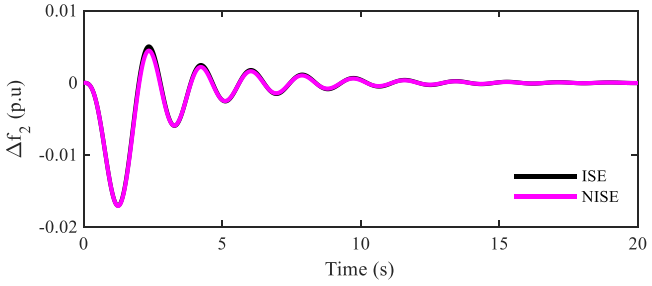


Fig. 16. Interconnected power system area 2 frequency deviations with two cost functions.

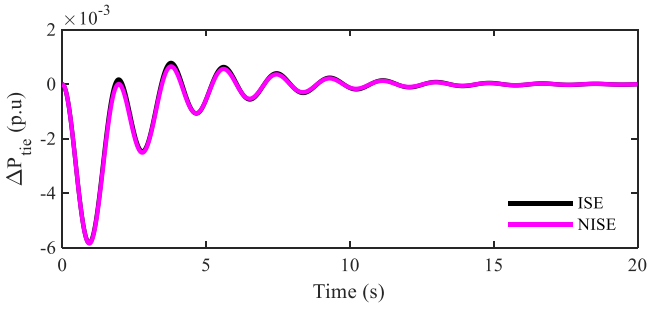


Fig. 17. Interconnected power system tie-line power deviations with two cost functions.

Investigation of proposed cost function and controller performance during random load changes in presence of noise

In this case, a random load fluctuation and noise are taken into account to evaluate the effectiveness of the suggested controller, whose ideal parameter gains are attained by a new cost function. Fig. 13 shows the load perturbations simulated randomly and the corresponding system frequency changes are presented in Fig. 14 for all the three controllers investigated in the case study 5.5. The observations reveal the same merits even the controller tuned values are corresponding to load change of 5%.

Upon getting satisfactory results using proposed cost function and controller in case of isolated power system, study is extended to multi area interconnected power system.

Table 3
Optimal parameter gains of PID and I-TPFOD controllers using SBO in system centralized mechanism.

Controller	k_p	k_i	k_d	n	β
PID	0.9999	0.9991	0.8921	–	–
I-TPFOD	0.8267	0.9699	0.9666	-0.4264	0.4631

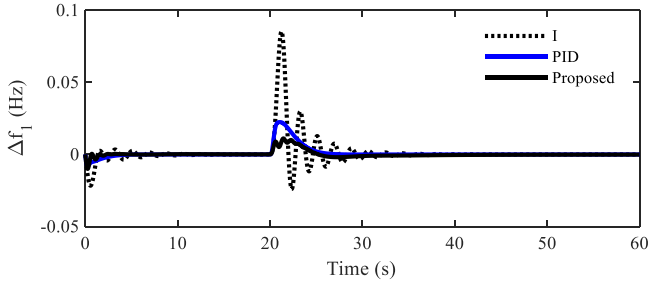


Fig. 18. . Δf_1 profiles with various controllers of interconnected power system in system centralized mechanism.

Study of the proposed cost function on interconnected power system

The proposed cost function is tested on two area interconnected power system with area centralized and system centralized mechanisms. In case of system centralized mechanism, the integral gain parameter searched by SBO is 0.6532 with ISE cost function. Using NISE, it is 0.62082. This solution varies based on the selection of threshold parameter. However, satisfactory results are identified irrespective of its selection. With these tuned integral controller values, simulations are carried out at 1% load change and corresponding results are provided in Figs. 15, 16 and 17. In individual graphs, the responses achieved with ISE and NISE are looks similar reveal the merits of the algorithm. Further, significant improvements are possible with suitable threshold parameter. In case of area centralized mechanism, the controller gains of area 1 and area 2 are 0.84005 and − 0.10189 extracted from solution region using ISE cost function. With NISE cost function, 0.79555 and − 0.084547 are identified as optimal integral controller gains. The results carry continuous improvements with these new controller gains.

Simulation results of interconnected power system

Based on case studies investigated on both systems, the proposed controller is used as supplementary controller in AGC to improve the frequency profile. In case of system centralized mechanism, one controller provides control calls to all the areas during load perturbations to minimize the power imbalance. The optimal controller gains of the proposed controller are presented in Table 3. For comparisons, integral and PID controllers are opted with SBO tuning. For this case, two different loadings (1% and 5%) are considered initiated at 0 and 20 s. The changes in area 1 and area 2 frequencies are presented in Figs. 18 and 19. Fig. 20 shows the inter area tie line power changes subjected to load

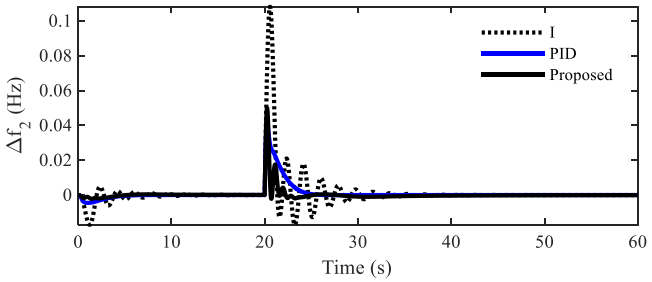


Fig. 19. . Δf_2 profiles with various controllers of interconnected power system in system centralized mechanism.

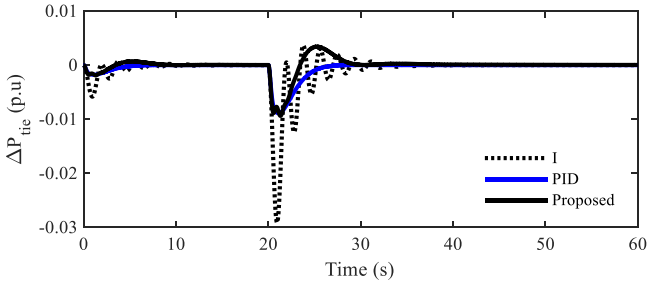


Fig. 20. . ΔP_{tie} profiles with various controllers of interconnected power system in system centralized mechanism.

perturbations. In case of area centralized mode, separate controllers provided for each area to give control commands to minimize the frequency and tie line power deviations of the interconnected power system. Table 4 reported optimal parameter gains of the PID and proposed controllers. The comparative results are presented in Figs. 21, 22 and 23. In both scenarios, proposed controller produces better results, and all these improvements are easily observable from simulation results.

Simulation results of the interconnected power system with renewable units

The wind power plant is integrated in the area-1 to test the performance of the classical, fractional, and proposed controllers in presence of load perturbations and wind intermittency. The frequency changes of the both areas and tie-line power disturbances are presented in Figs. 24, 25 and 26. The optimal controller gains identified for conventional systems are not adaptable for non-linear systems as the response of the FOPID controller is not acceptable as shown from Fig. 24 to Fig. 26 unlike Fig. 11. However, the proposed controller significantly improves the performance metrices compared to integral, PID and FOPID controllers. This is the additional benefit achieved with the I-TPFOD controller.

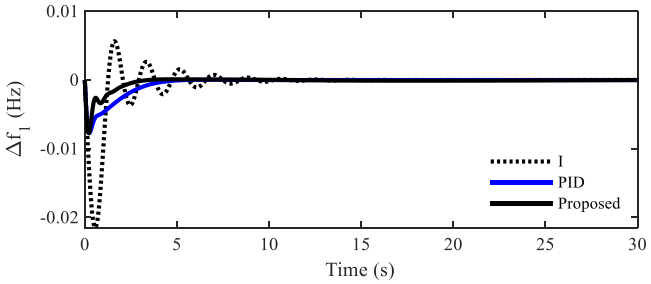


Fig. 21. . Δf_1 profiles with various controllers of interconnected power system in area centralized mechanism.

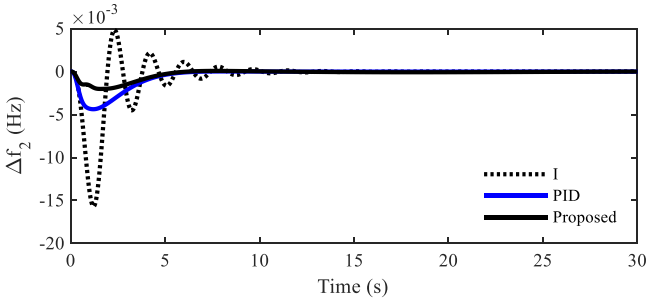


Fig. 22. . Δf_2 profiles with various controllers of interconnected power system in area centralized mechanism.

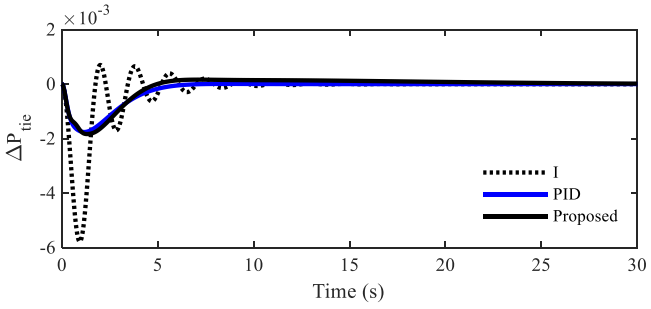


Fig. 23. . ΔP_{tie} profiles with various controllers of interconnected power system in area centralized mechanism.

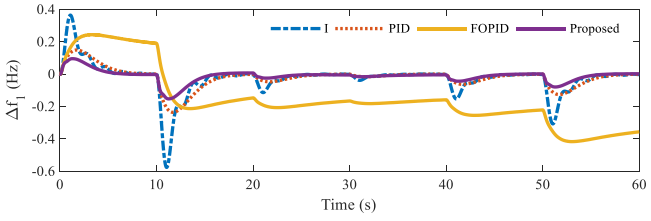


Fig. 24. Δf_1 profiles with various controllers of interconnected power system in presence of wind.

Table 4
Optimal parameter gains of PID and I-TPFOD controllers using SBO in area centralized mechanism.

Controller	k_{p1}	k_{i1}	k_{d1}	$n1$	$\beta1$	k_{p2}	k_{i2}	k_{d2}	$n2$	$\beta2$
PID	0.9997	0.9999	0.9995	–	–	0.1438	0.0623	0.6853	–	–
I-TPFOD	0.9864	0.9274	0.8406	-0.1770	0.7352	0.1528	0.3104	0.7376	-0.7536	0.5358

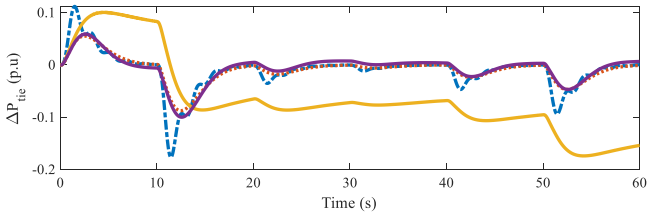


Fig. 25. ΔP_{tie} profiles with various controllers of interconnected power system in presence of wind.

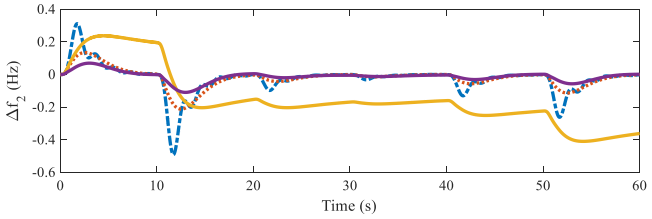


Fig. 26. . Δf_2 profiles with various controllers of interconnected power system in presence of wind.

Conclusions

In order to determine the best gain parameters for the additional controllers of the AGC, a new cost function is proposed in this research. The suggested cost function can deliver superior results to ISE with less samples in isolated, connected, and renewable integrated systems. To implement this new cost function, no additional information is required since it is framed with available signals used by traditional ISE function. Further, I-TPFOD controller is proposed in this paper and comparisons are made with integral, PID and FOPID controllers to extract the merits of the approach. The proposed cost functions and controller are tested on isolated power system at different loadings, sampling intervals and noise. Furthermore, investigations are extended to interconnected power system scenarios with proposed cost function and controller to extract the additional benefits. Overall, the performance measure outperforms ISE in terms of results. The proposed controller is adaptable for isolated, interconnected, and renewable integrated systems and produces significant improved results compared to classical controllers.

Author statements

No organization had funded this work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

1. Test system parameters: $T_g = T_{g1} = T_{g2} = 0.08$; $T_t = T_{t1} = T_{t2} = 0.3$; $R = R_1 = R_2 = 2.4$; $B_1 = B_2 = 0.425$; $K_p = 120$; $T_p = 20$(Instead of H and D, power system model parameters expressed as K_p and T_p [38]); $P_s = 0.545$.
2. SBO parameters: NB = 100; Iterations = 50; $\alpha = 0.1$; mutation probability = 0.05; Spread factor = 0.02.

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